

HUMAN-CENTERED VLA ROBOT SAFETY



Safe Vision-Language-Action Policy

$$\pi_{\theta} : (vision, language) \rightarrow action$$

VLA to **semantically** ensure **safe actions** in alignment **with human values**



Human-Centered Semantic Action Safety



Human Well-being

Protect human from physical harm and injuries.



Environmental Integrity

Preserve human assets and their environment structures.



Social Compliance

Respect laws, regulations, etiquettes and social values.

Design-Time Safety



Alignment: Learning to recognize and act in safety-critical scenarios



Data coverage: Safety-critical data sampling and scaling



Deployment-Time Safety

Action Safeguards: Constraining runtime unsafe action by selection and steering



OOD Safety: Detect, act and fallback under uncertainty and OOD



Validation-Time Safety

Safety Benchmarks: Standardized safety evaluation suites



Robustness Stress-testing: Validating across perturbations and edge cases